



Adductor tendinopathy as a source of ongoing chronic postoperative inguinal pain—evaluating outcomes of adductor tenotomy

T. W. K. van der Meer^{1,2} · M. J. A. Loos^{1,2} · J. van Grinsven^{1,2} · M. R. M. Scheltinga^{1,2} · W. A. R. Zwaans^{1,2,3}

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Abstract

Purpose To evaluate the outcome of an adductor muscle tenotomy in patients with chronic postoperative inguinal pain (CPIP) with a therapy-resistant adductor tendinopathy.

Methods In this retrospective cohort study, CPIP patients who underwent an adductor tenotomy between January 2019 and December 2024 at MMC were identified using a CTcuc search and an operative code. They were categorised according to the extent of surgical procedure (isolated group: adductor tenotomy only; combined group: mesh removal and/or tailored neurectomy combined with adductor tenotomy). Outcome was successful (excellent or good) or not (moderate or poor).

Results A total of 32 tenotomies were identified in 29 CPIP patients (25 males, mean age 59 years). All had previously undergone conservative therapies without achieving significant pain relief. The success rate in the isolated group was 57.1% (8 of 14 procedures). Success in the combined group was attained by 61.1% (11 of 18 procedures). No major complications were reported.

Conclusions Adductor tendinopathy can contribute to symptoms in some patients with CPIP. An adductor muscle tenotomy may be effective in approximately 60% of CPIP patients with a therapy-resistant adductor tendinopathy, either as an isolated procedure or as an adjunct during mesh removal and/or tailored neurectomy. These preliminary findings call for the need of further studies with standardised diagnostic criteria and validated outcome measures.

Keywords Chronic postoperative inguinal pain · CPIP · Adductor tenotomy · CPIP treatment · Adductor tendinopathy · Adductor tendinitis

Introduction

Inguinal hernia repair is one of the most frequently performed surgical procedures worldwide. Despite a variety of available operative techniques, the global gold standard is mesh-based repair, due to its lower recurrence rates compared to non-mesh alternatives [1]. However, approximately 10–12% of patients develop chronic postoperative inguinal

pain (CPIP), defined as groin pain persisting > 3 months following surgery. This pain impairs daily activities or work capacity in up to 6% of patients undergoing an inguinal hernia operation [2, 3].

The aetiology of CPIP is multifactorial and incompletely elucidated [4]. Contributing factors are chronic inflammatory reactions of structures in proximity to the mesh, nerve entrapment or iatrogenic tissue injury [5]. When conservative treatments including pharmacological treatment and physical therapy fail to alleviate symptoms, surgical interventions such as tailored or triple neurectomy, mesh removal or a combined procedure are considered [2, 6–10]. While approximately two-thirds of these patients experience pain relief following these procedures, a subset of approximately 30% of patients continues to suffer from persisting CPIP, potentially due to alternative sources of pain [2, 11]. In these patients with (refractory) CPIP, adductor muscle pathology may emerge as a significant contributor [12].

✉ T. W. K. van der Meer
T.vanderMeer@mmc.nl

¹ Department of General Surgery, Máxima Medical Centre, Veldhoven, Netherlands

² Centre of Expertise for Complex Groin and Abdominal Wall Pathology, SolviMáx, Eindhoven, Netherlands

³ NUTRIM School of Nutrition and Translational Research in Metabolism, Maastricht University Medical Centre, Maastricht, Netherlands

Clinical signs associated with adductor tendinopathy are local pain following provocative adduction of the upper leg or pain upon compression of the adductor tendon [13–16]. It has been hypothesised that mesh-induced inflammation may extend to the origin of the adductor tendon, leading to this chronic tendinopathy [17]. While conservative treatments remain the treatment of choice, adductor tenotomy may be considered as a last-resort therapeutic option. Precedent for this procedure exist in the athletic population for chronic adductor-related groin pain [18–22], although direct comparability to the CPIP population is limited by differences in aetiology and patient characteristics. This study is the first to determine the efficacy of this procedure, either as an isolated option or as an adjunct to mesh removal and/or tailored neurectomy in CPIP patients with a therapy-resistant adductor tendinopathy.

Methods

Study characteristics

This retrospective cohort study was conducted in the Department of General Surgery, Máxima Medical Centre (MMC), the Netherlands. MMC is a tertiary referral centre specialising in complex abdominal wall and groin pathology. SolviMáx, an integrated institution within MMC, focuses on the assessment and treatment of these patients, primarily referred from other hospitals, with an annual referral volume between 1000 and 1200 cases.

This study evaluates two patient groups with CPIP following inguinal hernia repair who received an adductor tenotomy, either as an isolated option or as an adjunct to mesh removal and/or tailored neurectomy between January 2019 and December 2024. Patients were identified through two complementary methods. An electronic health record search (CTcue) with predefined terms in combination with SolviMáx entry was queried. Moreover, each adductor tenotomy procedure is tabulated using a dedicated surgical code. Data of possible eligible patients were entered in a separate database for further analysis.

The study was conducted according to the Declaration of Helsinki. The Zuyd-Máx medical ethics committee judged that evaluation of the study protocol was not required since this effort was considered an evaluation of patient outcome and auditing of own results (METC number: N25.092). In addition, each patient who is evaluated in our SolviMáx department standardly provides written consent for the future use of data for research purposes.

Eligibility criteria participants

Inclusion criteria were age ≥ 18 years, a history of open or laparoscopic mesh-based inguinal hernia repair, diagnosis

of CPIP (defined as persisting pain > 3 months following repair)[2], clinical signs consistent with adductor tendinopathy (i.e. at least one positive finding including pain on resisted adduction of the upper leg, pain at the origin of the adductor magnus at the pubic bone, or pain upon compression of the adductor magnus muscle)[13–16], adductor tenotomy, and availability of an operative report. Prior to tenotomy, patients had undergone a combination of conservative treatment options, including several analgesic medications, repeated injections with local anaesthetic agents, core stability training (CSO), or pulsed radiofrequency (PRF). Patients were excluded if groin pain was attributable to causes other than CPIP (e.g., recurrent hernia, hip joint pathology, meshoma, genitourinary pathology or gynaecological pathology) as determined by clinical assessment and imaging when applicable. Neuropathic causes of CPIP were deemed less likely in the absence of positive or negative sensory symptoms [23]. No standardised algorithm or formal threshold was utilised. Selection for tenotomy was individualised and based on the treating surgeon's expertise.

Data accrual

Electronic health records were used for extracting demographics, age, gender, height, weight, body mass index (BMI) and surgical history. In addition, operative details, type of anaesthesia, intraoperative complications, postoperative complications, follow-up time and patient satisfaction were collected. Follow-up was performed 6–8 weeks postoperatively by the surgeon who performed the operation. The follow-up was extended until a satisfactory result was attained, or when additional surgical consultation was deemed non-meaningful.

Preoperative imaging

Preoperative imaging, such as MRI or ultrasound, was not standardised in the diagnostic work-up for CPIP-related adductor tendinopathy during the study period. The diagnosis was based upon patient history and physical examination, using criteria validated in the athletic population [13, 14, 16]. However, in some cases, imaging had nonetheless been performed. These findings were extracted from the electronic patient records. If the indication explicitly addressed adductor or pubic pathology and was performed in a reasonable time to the procedure it was considered part of the diagnostic stratagem. Imaging performed for unrelated indications or a > 2 year interval from surgery were not included.

Specifics of adductor tenotomy

Prior to the surgical procedure, patients were counselled on the pros and cons and potential risks associated with the

surgical procedure. Potential complications that were discussed were wound infection, haematomas, and the likelihood of therapeutic failure. All patients provided verbal consent to the specifics of the operation.

The procedure was performed in day care under general anaesthesia, or occasionally spinal anaesthesia. Patients were positioned supine with the affected limb in abduction (Fig. 1). A 2–3 cm transverse incision was made over the adductor tendon 1–2 cm proximal to the level of the inguinal fold. The tendon was freed from its surroundings and ligated distally, followed by transection using cautery, some 1–2 cm from its origin at the pubic bone (Fig. 2 and 3). A complete adductor tenotomy was performed, as opposed to a selective partial release where only the pathological part of the tendon is released [24]. Tendon release was confirmed by passive alternating hip adduction and abduction performed by the surgical team (Fig. 4). The wound was closed in two layers and infiltrated with 10 cc of bupivacaine. Patients were advised to avoid large strides for four weeks.



Fig. 1 Patient positioned supine with affected limb in abduction



Fig. 2 Adductor tendon dissected



Fig. 3 Adductor tendon ligated distally followed by transection using cautery, some 1–2 cm from its origin at the pubic bone



Fig. 4 No tendon structures visible

The decision to concurrently perform a mesh removal and/or a tailored or triple neurectomy depended on patient history, physical examination, and previous response to a range of earlier treatments, including analgesic injections and PRF. Mesh removal procedures were performed in the case of mesh-related pathology that was not previously addressed, according to a previously published protocol [9]. In cases of pre-operative sensory disturbances or intraoperative identification of an entrapped inguinal nerve (iliohypogastric, ilioinguinal, or genitofemoral), the nerve was excised via a separate excision following a tailored approach.

Outcome

Outcome was patient satisfaction at end of follow-up, without additional therapies. A four-point verbal rating scale was used, based on the electronic patient record (EPR) surgical

outpatient interpretation (1. Excellent, I am pain-free, 2. Good, I am almost pain free, 3. Moderate, although there is some pain reduction, I am still frequently bothered by pain complaints, 4. Poor, the operation had no effect, and the pain is virtually the same or worse) [25]. The outcome was dichotomised into successful (Excellent/Good) and unsuccessful (Moderate/Poor).

Data analysis

Data were analysed using IBM SPSS version 29 software (SPSS Inc., Chicago, IL, US) and were expressed as means ($\pm$ standard deviation), medians (range) or percentages (n) as appropriate. The outcome measure and complication rates were calculated using descriptive statistics. Each operation was analysed independently to allow determination of the effect of adductor tenotomy. In the case of bilateral adductor tenotomy, each side was evaluated separately. Given the small sample size and exploratory nature of this study, inferential statistics usage was limited. Fisher's exact test was performed to compare the proportion of favourable outcomes between the isolated and combined groups, and between shorter (<6 months, $n=22$) and longer (≥ 6 months, $n=10$) follow-up periods. A two-tailed p -value <0.05 was considered statistically significant.

Results

Demographics and operative specifics

A patient flowchart is depicted in Fig. 5. A total of 209 patients with adductor tendinopathy were identified, 178 (85%) had not undergone tenotomy and were treated conservatively, 32 adductor tenotomies in 29 patients fulfilled study criteria. Of these, 14 were isolated tenotomies, whereas 18 were combined with mesh removal ($n=3$), tailored neurectomy ($n=8$), or both ($n=7$). In the isolated group, 78.6% and 21.4% of patients had previously undergone neurectomy and mesh removal at the affected side, respectively, compared with 21.4% and 11.1% in the combined group.

Patients' characteristics are depicted in Table 1. Operation time of the isolated tenotomy was 15 min (median) and 64 min in the combined group.

Preoperative imaging

Preoperative imaging was conducted in 12 of 29 patients (41.4%), primarily for reasons unrelated to adductor tendinopathy, such as recurrent hernia and mesh complications. Notably, in 3 patients (10.3%), MRI and/or ultrasound were performed to investigate potential abnormalities related to

adductor tendinopathy. However, no abnormalities were found on imaging, despite a clinical diagnosis of adductor tendinopathy.

Outcome

Outcome is depicted in Fig. 6. At the 4.8 months follow-up [median, 2–19], outcome after an isolated tenotomy was excellent in 5 procedures (35.7%) and good in 3 procedures (21.4%), leading to success in 8 procedures (57.1%). As the outcome was moderate in 5 procedures (35.7%) and poor in one (7.1%), the outcome was unsuccessful in 6 procedures (42.9%).

In the combined tenotomy group, outcome after 4.2 months follow-up [median, 1–60] was excellent in 5 procedures (27.8%) and good in 6 procedures (33.3%), resulting in success in 11 procedures (61.1%). As the outcome was moderate after 4 procedures (22.2%) and poor in 3 procedures (16.7%), the outcome was unsuccessful in 7 procedures (38.9%). No statistically significant difference in favourable outcomes was observed between the isolated (57.1%) and combined group (61.1%), Fisher's exact test ($p=1.000$). Regarding follow-up, one redo tenotomy for recurrent adductor tendinopathy was performed in the isolated group after 8 months, and one in the combined group after 60 months.

No statistically significant difference in favourable outcome was observed between a shorter (<6 months, $n=22$) and longer (≥ 6 months, $n=10$) follow-up period (Fisher's exact test, $p=0.467$).

Complications

No major complications were observed. Minor surgical complications were haematoma ($n=2$, 1 in each group) that was managed conservatively.

Discussion

Chronic postoperative inguinal pain (CPIP) is a multifactorial and incompletely elucidated complication affecting 10–12% of patients undergoing inguinal hernia repair [2, 3]. A previous study showed that musculoskeletal pathologies should be recognised in CPIP patients [12], and it has been hypothesised that mesh-induced inflammation may affect surrounding musculoskeletal structures, including the origin of the adductor tendon, potentially leading to chronic adductor tendinopathy [17]. However, this hypothesis remains speculative due to lack of substantial evidence linking mesh-induced inflammation directly to the development of adductor tendinopathy. Further research is needed

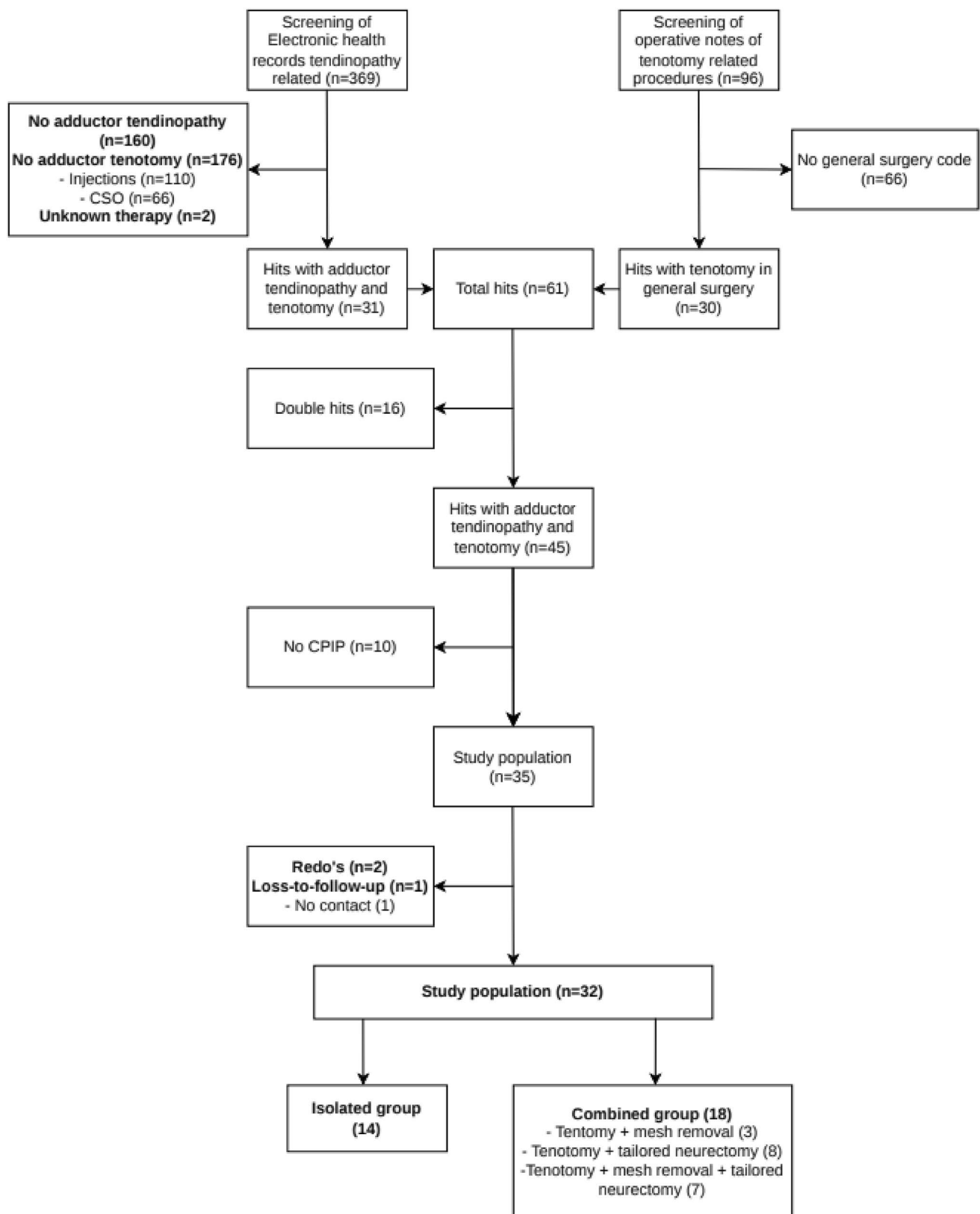


Fig. 5 Flowchart of patients undergoing adductor tenotomy for CPIP, CSO Core stability exercises

Table 1 Characteristics of CPIP patients (n=29) undergoing adductor tenotomy (n=32), either isolated or combined with mesh removal and/or tailored neurectomy

Adductor tenotomy	Isolated (n=14)	Combined with mesh removal and/or tailored neurectomy (n=18)
Age (years)	58±11.3	59±12.4
Sex ratio (male:female)	13:1	15:3
Body Mass index (kg/m ²)	26.6±2.5	27.9±3.4
History of hernia repair, n (%)		
Open/Lichtenstein hernia repair	6 (42.9)	9 (50)
Laparoscopic hernia repair	4 (28.6)	6 (33.3)
Open and laparoscopic hernia repair	4 (28.6)	3 (16.7)
Side of hernia, n (%)		
Left	10 (71.4)	5 (27.8)
Right	4 (28.6)	13 (72.2)
Follow-up (months)	4.8 [2–19]	4.2 [1–60]
Operation time (min)	15 [8–28]	64 [27–118]
Anaesthesia, n (%)		
General	13 (92.9)	17 (94.4)
Spinal	1 (7.1)	1 (5.6)
Mesh removal, n (%)	-	10 (55.6)
Neurectomy, n (%)		
Ilioinguinal nerve	-	14 (43.8)
Iliohypogastric nerve	-	7 (22.6)
Genitofemoral nerve	-	2 (6.3)
Complications, n (%)		
Haematoma, conservative	1 (7.1)	1 (5.6)
Previous neurectomy at affected side n (%) [*]	11 (78.6)	4 (22.2)
Previous mesh removal at affected side n (%) [*]	3 (21.4)	2 (11.1)

Mean ± standard deviation, or median [range]

^{*}Previous neurectomy or mesh removal on affected side refers to a distinct surgical procedure performed prior to the tenotomy, and is independent of the isolated or combined classification

to elucidate this proposed mechanism and establish a clearer relationship between a mesh implant and subsequent musculoskeletal complications in CPIP patients.

Multiple studies on algorithms for intractable CPIP have mainly focused on neurectomy and/or mesh removal but fail to mention the role of adductor tendinopathy [2, 6–10, 26]. A multifactorial surgical approach, including neurectomy, partial tenotomy of the rectus and adductor longus muscle, and posterior hernia mesh repair, has been described for pubic inguinal pain syndrome (PIPS) [27]. To date, such a multifactorial approach, including tenotomy, has not been described in CPIP patients. Aim of the present study was to determine the efficacy of a tenotomy, either as an isolated procedure or as an adjunct to mesh removal and/or tailored neurectomy, reflecting the clinical reality that CPIP is often multifactorial. The results suggest that CPIP in some patients may in part be related to symptomatic adductor

tendinopathy. Conservative treatment should remain the first-line approach for CPIP, however, in this exploratory study, adductor tenotomy was effective in approximately 60% of CPIP patients with therapy-resistant adductor tendinopathy, supporting its consideration as an additional therapy option.

Detailed history taking is crucial in CPIP. The syndrome should be considered as a spectrum entailing neuropathic, nociceptive, nociplastic or combined forms [4, 6]. Neuropathic CPIP pain is often described as a sharp, burning or shooting painful sensation ('pins and needles') that is progressive after stimulation. In contrast, nociceptive CPIP pain is mostly reported as throbbing or nagging [4, 13]. Based on clinical history from the EPR in the present cohort, patients predominantly displayed a nociceptive pain pattern. In our clinical experience, pain originating from adductor tendinopathy is characteristically triggered by lifting the affected leg when climbing stairs, crossing the legs while sitting, or turning in bed, with some patients reporting waking up due to their familiar pain. Getting in and out of a car is often painful. These specific symptoms can help to differentiate between the different CPIP entities and may guide patient selection for a tailored surgical approach.

Physical examination is also essential in diagnosing an adductor tendinopathy and identifying the various CPIP entities. Tests such as the adductor squeeze test, pain on resisted leg adduction, and pubic bone palpation are widely used in the athletic population to evaluate whether an adductor tendinopathy is present [13–16]. Clinical signs consistent with adductor tendinopathy were required for inclusion, therefore, all patients in our non-athletic cohort showed at least one positive test. Although validated in sports medicine, their data in our non-athletic CPIP cohort was clinically informative, underlining the importance of physical examination in the diagnostic evaluation of CPIP patients.

Imaging techniques like MRI and ultrasound show limited diagnostic value for adductor tendinopathy. MRI and musculoskeletal ultrasound can assess tendon structure, tears, and differentiate adductor tendinopathy from other pathologies. Imaging may demonstrate features like tendon thickening, increased tendon signal intensity and tendon bulging [28], but there is often a poor correlation between imaging findings and symptoms, as asymptomatic individuals may also demonstrate these tendinopathic changes. This has been demonstrated in our own cohort, in 3 patients (10.3%), imaging was performed but all results were unremarkable despite clinical diagnoses. This supports the notion that imaging for adductor tendinopathy often lacks clinical relevance. Most studies are performed in athletic populations [13, 29, 30], and the absence of standardised protocols in our cohort further restricts conclusions. Therefore, patient history and physical examination

Fig. 6 Satisfaction of CPIP patients after adductor tenotomy

remain pivotal for diagnosis, emphasising the need for more research on imaging's role in this patient population.

A complete adductor tenotomy was performed in this study, which has traditionally shown good functional results in athletes with chronic groin pain, despite some reduction in adductor strength [31]. Selective partial adductor release has been proposed as an alternative to prevent loss of strength in athletes [24]. However, this cohort was older, non-athletic and treated for pain relief. Therefore complete tenotomy was considered an appropriate treatment approach. Adductor strength and function were not systematically assessed, future studies should include such outcome measures. Recurrence after complete tenotomy has been described due to the body's natural healing process, where the ligated tendon scars back to the pubic bone [32]. This provides a possible explanation for the two redos in our cohort.

The present study has several limitations. All patients had undergone conservative treatments, and most had undergone prior surgeries before the tenotomy, including tailored neurectomy and mesh removal, which may confound outcomes. Patients may also not always be able to distinguish tendinous from inguinal canal pain. Patient selection for tenotomy and subsequently, indication for isolated or combined (neurectomy and/or mesh removal) procedure was based on the surgeon's expert judgement rather than a standardised algorithm. The retrospective design and the reliance on a single data analyst introduce the risk of selection and information bias, follow-up assessments were not blinded, and systematic physical examination was absent. Given the small sample size and exploratory nature of this study, inferential statistics were performed selectively. Fisher's exact test showed no significant difference in favourable outcomes between the isolated and combined groups (57.1% vs

61.1%, $p=1.000$) or between shorter (<6 months) and longer (≥ 6 months) follow-up ($p=0.467$) periods. Both results should be interpreted cautiously due to the limited statistical power. The absence of a control group and natural history comparison limits the interpretation of the results, and the sample size was too small to identify predictive factors for success. Outcome was based on a four-point satisfaction scale, quantitative pain scale, symptom duration, and objective assessment of adductor strength and function was not systematically recorded. Imaging was not part of the standardised diagnostic work-up, in case it was available, it did not follow a standardised protocol. Despite these limitations, the present study generates a hypothesis regarding the role of adductor tenotomy in CPIP patients with clinical adductor tendinopathy, calling for future studies.

In conclusion, this is the first study to report that adductor tenotomy may be considered in a specific group of CPIP patients with adductor tendinopathy failing conservative treatment, either as an isolated procedure or as an adjunct to mesh removal and/or tailored neurectomy. Future studies with standardised diagnostic and selection criteria, validated outcome measures, including objective strength and functional adductor assessment, standardised imaging, and adequate follow-up are needed to confirm these preliminary findings and define the role of adductor tenotomy in CPIP treatment.

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Declarations

Conflict of interest T.W.K. van der Meer, M.J.A. Loos, J. van Grinsven, M.R.M. Scheltinga and W.A.R. Zwaans declare that they have no conflict of interest.

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